BARREL + ENDCAP



HELIX · RUN-3 LLP · MSVtx

Does a detector event have a grammar?

LEXICON mk_1: the goals, what has run (R0–R15), and what is registered next

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LEXICON in one slide

1 The question

Quantize each detector feature, so an event becomes a **set of tokens**. Language models learn structure from which words keep company. Do detector tokens have structure of their own (a **grammar**), and is it useful?

Setting: Run-3 LLP displaced vertices in the muon spectrometer (MSVtx), barrel and endcap. Two networks, NN1 and NN2, on disjoint inputs form the ABCD plane; signal is HSS MC, background is HTmiss-low data.

2 Why it would matter

- A background-only token score is a candidate **ABCD axis**: its model never sees a signal label (only a one-bit sign orients it)
- Token dependence across the NN1/NN2 inputs could **predict closure** and propose the feature split
- Token embeddings as **classifier inputs**; structure that follows the decay chain would show what the nets learn

3 Where it stands

- Background tokens predict each other strongly; tokenizing is cheap at the right bin count
- Per-net token axes can close where the NN pair does not, but the best plane was found post hoc
- Rules frozen and five predictions registered (19 Oct); **no confirmation run yet**

Part I · Goals and experiments
the three goals, every proposed experiment by goal, the theory and rules behind them

→

Part II · What has run
R0–R15 on base_mk1: tokens, density models, the leak, per-net axes (exploratory)

→

Part III · Plan
registered predictions P1–P5, then the runs each goal needs, then open decisions

PART I

Goals and experiments

What LEXICON is for, every experiment proposed for it, and the theory behind them

PROPOSED

LEXICON asks: does a detector event have a grammar, and can we use it?

1

Is there a grammar?

Across a wide set of detector features, not just the networks' inputs, do background events follow rules about which features show up together? And do those rules hold however we bin the data, split it, or choose the features?

2

Does signal speak differently?

Does a signal event break background's rules, in how its features go together and not just in their values? Can that tell signal from background?

3

Can the grammar build a better ABCD plane?

3a. A score built only from background's grammar, which never sees a signal label, could serve as one axis of the plane.

3b. The rules that link the two networks' inputs could show which features each network should get, so the two axes stay independent. We'd then check that split against GAMBIT's.

PLAIN WORDS

BERT ideas, GPT ideas, and synergistic features

BERT Fill in the blank

Hide one token; guess it from **all the others**.

nTracklet·3 **hits·?** MSeg·2

Gives a rich summary of each event and a quick "how unusual?" score. **Can't** give a true probability for the whole event.

ran	CutFormer and counts: all of Part II
1.5, 1.19	attention test; how much to hide
2.3, 2.11–13	signal transformer; token classifiers
2.16	token summaries into NN1/NN2

GPT Predict the next one

Read tokens in an order; predict each from **the ones before**.

nTracklet·3 → hits·7 → **?**

Gives an exact probability for the whole event, so it can **measure** the grammar and compare signal with background. **Catch:** features have no natural order, so try many.

1.1–1.4	exact models; best order; size
2.6–2.10	sig/bkg ratio; decay chain
2.14	next token = signal or background
3b.1, 3b.4	shared info, NN1 vs NN2

NEW GROUND Synergistic features

Weak alone, **informative together**.

$$I(Y; X, Z) = I(Y; X) + I(Y; Z \mid X)$$

A separation ranking sees only $I(Y; Z)$. What Z adds is $I(Y; Z \mid X)$, which can be large when $I(Y; Z) = 0$.

Synergy: information only in the combination (XOR)
Context: changes what others mean (η, pileup, region)
Suppressor: cancels the noise in a good feature

Why only LEXICON can look: the NN inputs were picked for separation, which drops these first. LEXICON learns how features **co-occur**, so it can find them.

1.23	quality-only token pool (today: NN inputs only)
1.12–1.16	wide-pool search, feature-set study
1.20	class-conditional permutation tests
F13–F17	checks: detector or MC-only structure?

Like "the" and "of": no topic of their own, but they carry the syntax.

Both, head to head: 2.15 BERT vs GPT on the same data · 1.17 pretraining · 3a axes. **Neither:** latent classes and factors (1.7–1.9), Evt2Vec. **So far only BERT-style has run.**

PLAIN WORDS

What is a token in detector data?

1 What a token is

The smallest piece a language model reads. It never sees raw text: a **tokenizer** first chops the text into pieces from a fixed **vocabulary**, and the model learns which pieces keep company.

LEXICON asks the same of detector data. The catch: our features are **numbers**, not words, so we first have to invent the pieces.

2 Compared with LLMs

Text: *"unbelievable"*

letters — u | n | b | e | l | ...

fragments — un | believ | able — *what LLMs use*

words — unbelievable

Detector: *nTracklet = 4*

raw value — **4** *a number, not a symbol*

binned — 

token — nTracklet · bin 3 — *what LEXICON uses*

	LLM	LEXICON
token	a word fragment	a feature in one range
sentence	a line of text	one event: about 26–30 tokens
vocabulary	tens of thousands of fragments	about 260 (barrel) and 840 (endcap)
order	given by the text	**none**: an event is a set
empty slot	–	a **missing** token

3 How we do it

1. Look only at **background** events.
2. Cut each feature's range into bins that hold **equal numbers** of background events: 8 in the barrel, 32 in the endcap.
3. **Merge** any bin too rare to learn from (under 5 events).
4. Give a missing value **its own token**.
5. **Check** the tokens keep what the raw numbers knew: they lose only 3–6%.
6. Each event becomes a **set** of about 26–30 tokens, ready for the models.

PLAIN WORDS

Feature, token, event: the words we use



A feature is a slot in a form; the token is the word filled into that slot for this event.

A token is a value, not a feature

One feature's value in one event after binning: the pair (feature, level), like "nTracklet in bin 3".

- The **feature** is the column, nTracklet.
- Its **vocabulary** is its set of levels: for example 8 bins plus a missing token.
- A **token** is the level that feature took in a particular event.

An event is a set, not a sequence

Exactly one token per feature, or the missing token, so its length is fixed: about 26 tokens in the endcap and 30 in the barrel today. Text varies in length and comes with an order.

- **BERT-style** keeps the set: feature-ID embeddings, no position.
- **GPT-style** reads the set in an order *we choose*, because the chain rule needs one. The order is a modeling device, not a property of the event: several are tried and averaged, and whether one fits better is tested (1.2–1.3).

The event as modeled

The token set is the binned version of the features we chose to include, so it is lossy: about 3–6% of what the raw values carry, and only the features in the set. Per-network axes use one network's subset. The model then gives that set a probability or a surprise score.

A token is a feature's binned value in one event; an event is the set of its tokens, one per feature; the models learn which combinations of tokens background events tend to produce.

PLAIN WORDS

Why background only

1 It's what ABCD needs.

An ABCD axis has to be independent of the other axis on background, and it can't be tuned to the signal it's meant to find. A model trained only on background sees no labels and no signal hypothesis. Its score just measures "how unlike background is this event", which is the textbook anomaly-detection setup.

2 It avoids the MC-vs-data trap.

Signal is MC and background is data, and they're also split by construction at HTmiss = 40 GeV. Anything trained with both learns those differences as if they were physics. A background-only model can't.

3 Bin edges have to be label-free.

If signal helped place the edges, the tokenizer itself would carry label information into everything downstream.

4 Statistics.

The endcap has 34,970 background events, against 9,996 signal events spread over six mass points. A grammar needs many examples of the same language.

TECHNICAL

Tokenizer methods: tried and planned

A token is a discrete symbol for "this feature fell in this range": a pair (feature, level). An event becomes a **set** of tokens, one per feature, plus a *missing* token where a value is absent.

Four choices define a tokenizer (a "path")

Unit	what one token stands for	binned single feature (S4) used; cut-stage tokens considered; object- or track-level tokens not realistic with these files
Edges	where the bins fall	equal-background-mass quantiles T0 (8, 16, 32 bins) used, atom-aware (spikes and discrete values keep their own token); Bayesian Blocks T2; log-quantile is identical to quantile
Vocabulary	how tokens are grouped	one sub-vocabulary per feature (V1) used
Assignment	how a value maps to a token	hard bin index (A1) used

120 possible paths → 25 with binned features → 19 viable. 8 screened; the unit, vocabulary and assignment choices never varied. Beyond fixed bins, untried: learned codebooks (van den Oord 2017), continuous number embeddings (Golkar 2023), or no tokens at all (Leigh 2024); the raw-value control stands in for the last.

Experiments that build and test tokens

steps 0–2	Edges per fold on background only, Gate A, occupancy floor: wave-1 paths pass in both regions	done
step 3	Tokenization cost against the raw-value control, 4 barrel schemes: 8 bins loses 4.9%, 32 bins 31%	done
R5	Open top quantile bin hides the signal tail; Bayesian Blocks split it	done
F4, F5	Endcap cost per scheme; Bayesian Blocks through every model	Open check
F6	Tail rule for the top and bottom quantile bins	Needs decision
F8	The 15 unscreened paths: other units, vocabularies, assignments	Later
F9	One shared tokenization for every feature set; sentinels become missing tokens	Proposed
1.23	Tokenize the quality-only pool for the synergy search	Proposed
1.19	Masking design: a mask token distinct from the missing token	Proposed

Which runs used which tokenization

Round	What ran	Runs	Barrel				Endcap				Feature set	Models
			b8	b16	b32	T2	b8	b16	b32	T2		
R0	Screen: edges, Gate A, occupancy	6	3	3	3	3	3	3	3	3	all NN inputs	Gate A, occupancy
R0–R1	Step 3, wave 1	3	2						1		all NN inputs	indep · CutFormer · raw · Evt2Vec
R3–R4	Barrel training seeds 0, 1, 2	3	3								all NN inputs	same as step 3
R3	Evt2Vec sweep, 32 and 128 dims	16	2	2	2	2	2	2	2	2	all NN inputs	indep · Evt2Vec
R4	Tokenization cost vs scheme	3		1	1	1					all NN inputs	indep · CutFormer · raw
R7	Count-model ladder, orders 1–3	5	3			1			1		all NN inputs	counts
R8–R10	Surprise vs the networks (49 jobs)	6	2			2			3	2	all inputs (leaky)	indep · counts · CutFormer · raw
R11	Per-net surprise (D6)	6	3						3		per net	indep
R12	Label-free plane; feature removals	12	5						9		per net, post hoc	indep
R13	Per-net count models	15							15		per net	counts, orders 1–3
R14	Timing bridge; shared mass point	7							7		per net + timing	indep · counts
	Runs that used each tokenization	**82**	**23**	**6**	**6**	**9**	**5**	**5**	**44**	**7**		
PLANNED, NOT RUN												
F4, F5	Endcap cost per scheme; T2 everywhere	–					○	○		○	all NN inputs	CutFormer · raw
F6	Tail rule (new scheme)	–	+tail						+tail		all NN inputs	steps 0–3
1.1	Exact likelihood vs pseudo	–	○						○		per net	MADE · GRU · trees
1.23	Quality-only pool	–	○						○		about 64 / 69 features	steps 0–3, then discovery

b8 / b16 / b32 = equal-background-mass quantile bins (T0); T2 = Bayesian Blocks. Every run: binned single features, one vocabulary per feature, hard bin index; edges fitted per fold on background only. **Wave 1** = barrel b8, endcap b32: every per-net run (R11–R14) reuses those tokens. A cell counts the runs in that row that used the scheme; a screen or sweep run can cover several schemes, and the two label-free plane runs (R12) are counted in both regions.

Data: RUNLOG.csv, 82 runs (R0–R14, backfilled R15), base_mk1 Round_1, base seed 0, 5 folds · cell = runs that used that region and scheme

Runs by model family: BERT vs GPT

Round	What ran	Runs	Baseline	BERT-style · fill in the blank			GPT-style · next one	Other	None
			Independence	Pair / triple counts	CutFormer	Raw-value control	MADE · GRU · causal · tree	Evt2Vec	Screen only
R0	Screen: edges, Gate A, occupancy	6							●
R0–R1	Step 3, wave 1	3	●		●	●		●	
R3–R4	Barrel training seeds 0, 1, 2	3	●		●	●		●	
R3	Evt2Vec sweep	16	●					●	
R4	Tokenization cost vs scheme	3	●		●	●			
R7	Count-model ladder, orders 1–3	5	●	●					
R8–R10	Surprise vs the networks (49 jobs)	6	●	●	●	●			
R11	Per-net surprise (D6)	6	●						
R12	Label-free plane; feature removals	12	●						
R13	Per-net count models	15	●	●					
R14	Timing bridge; shared mass point	7	●	●					
	Runs that used each family	**82**	**76**	**33**	**15**	**15**	**0**	**22**	**6**
PLANNED, NOT RUN									
1.1–1.2	Exact vs pseudo-likelihood; order ensemble	–			○		○		
1.4	Grammar budget from exact trees	–					○		
2.6–2.10	Signal-vs-background ratio; decay chain	–					○		
2.14	Label as the next token	–					○		
2.11–2.13	Token probes; fine-tuned classifiers	–			○				
2.15	BERT vs GPT, head to head	–			○		○		

Every run so far is the baseline or BERT-style. No GPT-style model has run, so nothing yet measures the grammar exactly. Independence is exact, with no context: the floor for both.

Data: RUNLOG.csv, 82 runs (R0–R14), base_mk1 Round_1, base seed 0, 5 folds · a cell marks a family whose scores the run produced or compared; totals add the row's runs

BERT-STYLE

Using a token, BERT-style: fill in the blank

Once tokenized: the process

1. **Look up**
   each token becomes a vector: a token embedding plus a feature-ID embedding. No position: an event is a set
2. **Hide**
   replace one or a few tokens with a `[MASK]` token, kept distinct from the *missing* token
3. **Read both ways**
   transformer layers let every visible token attend to every other
4. **Guess**
   predict the hidden token from that feature's own vocabulary: $p_\theta(t_i \mid t_{\setminus i})$
5. **Learn**
   minimize the cross-entropy on the hidden tokens, on background only
6. **Score**
   hide one token at a time and add up: surprise $s = -\sum_i \log p_\theta(t_i \mid t_{\setminus i})$; or pool the token vectors into one event vector

On our data

Train	background only, base_mk1 folds, every event scored out of fold; for an ABCD axis, one network's features only (D6)	D6
Axis	per-network surprise S_k paired with the other network	3a
Directions	keep per-feature surprise terms instead of one sum, so signal stops cancelling	3a.10
Vectors	event vectors into a linear probe, a nearest-neighbour score, or NN1/NN2 as extra inputs	2.11, 2.16, 3a.11
Fine-tune	add a signal/background head after the data/MC audit	2.12
Design	mask rate tuned per network; replacements drawn from background marginals; endcap → barrel pretraining	1.17, 1.19

Where it stands: CutFormer ran on all features in both regions (wave 1). Per-network CutFormer has not run. In the barrel, simple pair/triple counts beat it. **Limit:** the score is a pseudo-likelihood: no exact probability, so it cannot measure the size of the grammar.

GPT-STYLE

Using a token, GPT-style: predict the next one

Once tokenized: the process

1. **Pick an order**
   the features have no natural order, so choose one, π, and later try several
2. **Look up**
   token embedding plus feature-ID, and its place in the chosen order
3. **Read forwards**
   each token sees only the ones before it (a causal mask)
4. **Guess the next**
   predict $p_\theta(t_{\pi(i)} \mid t_{\pi(<i)})$ at every position at once
5. **Learn**
   minimize the cross-entropy over all positions, on background only
6. **Score exactly**
   the chain rule multiplies them: $\log p(\mathbf{t}) = \sum_i \log p_\theta(t_{\pi(i)} \mid t_{\pi(<i)})$, exact for any order; average over orders

On our data

Size of the grammar	$\widehat{C} = \langle \log p_{\mathrm{AR}}(\mathbf{t}) - \sum_i \log p(t_i) \rangle$, averaged over held-out background	1.4
Order	does any order fit better than seed noise allows?	1.2–1.3
Exact axis	$-\log p(\mathbf{t})$ per network as an ABCD axis, against the pseudo-likelihood	1.1, 3a
Signal vs background	fit signal too: $\Lambda = \log p_S - \log p_B$, split into what tokens carry alone and what their relations add; decay-chain test	2.6–2.10
Feature split	shared information between the NN1 and NN2 input blocks	3b.1, 3b.4
Next token = label	append the class as the last token and predict it	2.14

Models: MADE (default), a causal or any-order CutFormer, a GRU, and Chow–Liu trees; background only, per network. **Where it stands:** none has run. Region order unsettled: barrel first (page, Addendum 4) or endcap first (mk2).

BARREL + ENDCAP

Good vs bad tokens: an information-theory scorecard

A good token…	Measured by	Where it stands
keeps the information	Gate A: $R = \dfrac{\hat{I}_{\text{held-out}}(Y;T)}{\hat{I}_{\text{held-out}}(Y;X)}$ held-out information $= H(Y) + \overline{\log_2 p(y \mid \cdot)}$	Locked. BAD below 0.90 (provisional); $R > 1$ allowed. **Uses labels**, and is undefined for a feature with no separation of its own (NO_SIGNAL). Barrel: 16 INDETERMINATE
keeps the context	Tokenization cost $\Delta = S_{\text{tok}} - S_{\text{raw}}$ in nats per event, background only, as a share of the raw gain	Measured: 4.9% (barrel, 8 bins), 3.3% (endcap, 32 bins, upper bound). **Label-free**, so it is the judge for weak features
can be estimated	Occupancy: every token seen in ≥ 5 training-background events. Plug-in entropy is biased by about $\frac{K-1}{2N}$ nats	Locked (merge-to-floor; the missing token exempt). Barrel at 32 bins has more tokens than training events
takes part in the grammar	Co-occurrence: $\mathrm{PMI}(a,b) = \log \frac{p(a,b)}{p(a)\,p(b)}$; per-feature gain over independence	Measured with count models. Features with no partner in any barrel seed (nTracklet, MET_dphi, nBOL) add no co-occurrence gain
doesn't hide the signal	Where signal lands among background's tokens; tail resolution	Seen once: 85% of signal hits_ntot in a token holding 17% of background. Background-only checks are blind to it
isn't an artifact	HTmiss leak gate (single-token AUC ≥ 0.995); data/MC audit; edges stable across folds and seeds	Open: F13–F16

Steps 0–3 test the first four. The last two need labels and the data/MC audit. For synergistic features the order matters: Gate A cannot score them, so the **label-free tokenization cost decides**.

Every proposed experiment: 82, sorted by goal

F — Foundations: tokens, code and checks every goal relies on

19 experiments

Next 3 · Open check 5 · Proposed 6 · Needs decision 3 · Gated 1 · Later 1

G1 — Is there a grammar?

23 experiments

Next 8 · Proposed 10 · Gated 2 · Later 3

G2 — Does signal speak differently?

19 experiments

Next 13 · Proposed 3 · Gated 2 · Later 1

G3a — A token score as an ABCD axis

14 experiments

Registered 5 · Next 2 · Open check 1 · Proposed 1 · Needs decision 1 · Gated 4

G3b — Which features each network should get

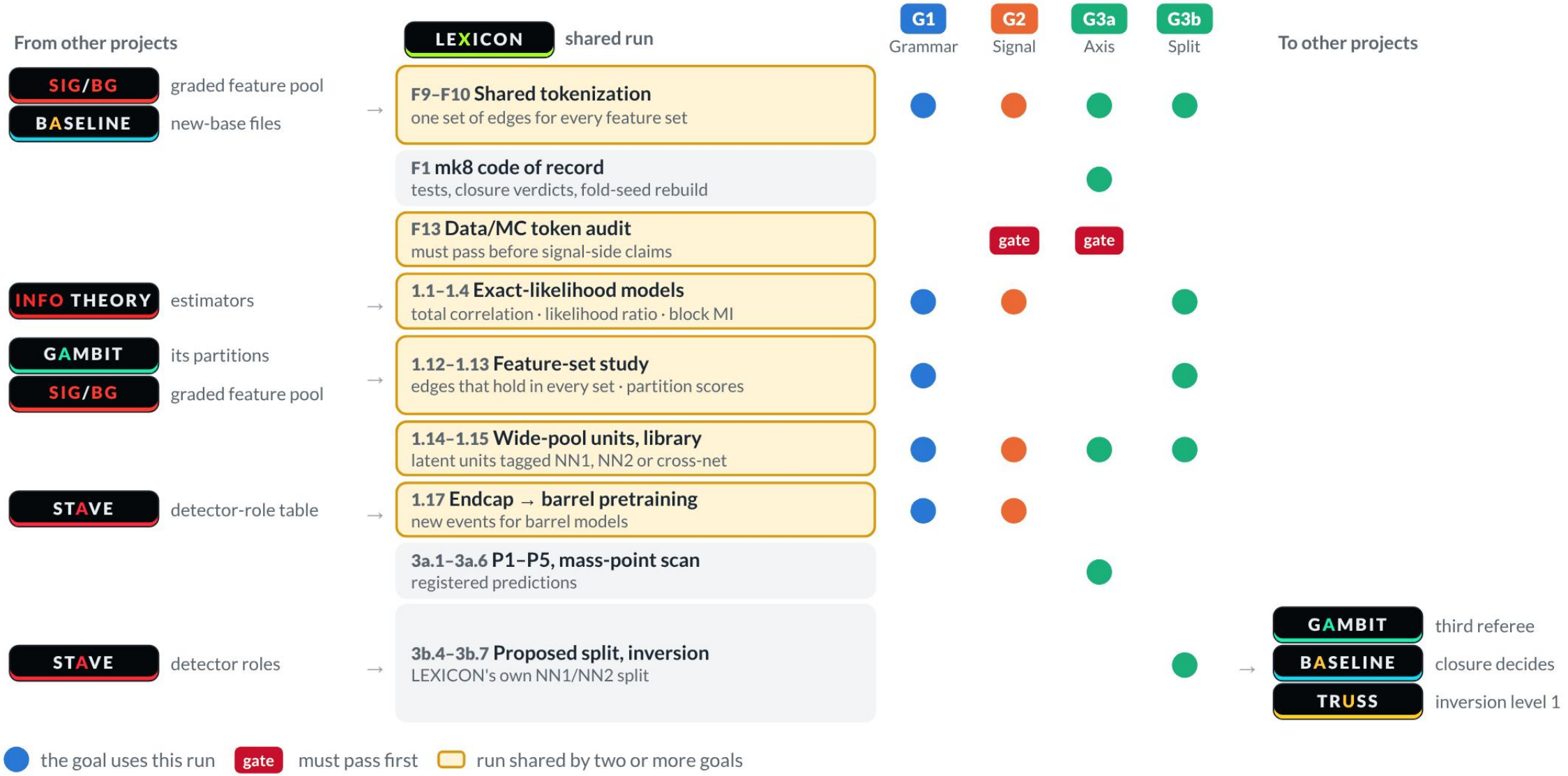
7 experiments

Proposed 4 · Gated 3

Status Registered · Next · Open check · Proposed · Needs decision · Gated · Later

Registered: criteria frozen before running (vetting v1) · **Next**: on the public timeline or the mk8 plan · **Open check**: vetting queue · **Proposed**: in a proposal, not adopted · **Needs decision**: waiting on an open decision · **Gated**: waits on another result · **Later**: deferred, optional or record-only

How the experiments overlap: one run, several goals

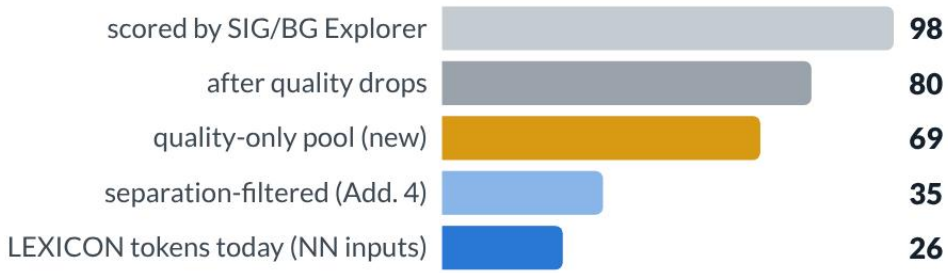


EXPERIMENTAL

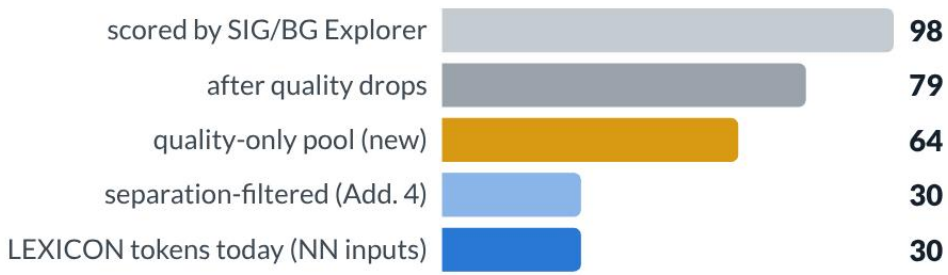
Creating synergistic features: plan of attack

How many features each choice keeps

Endcap



Barrel



SIG/BG Explorer mk5 (Prof. Johns' Sep-21 files). Quality drops: HTmiss-linked 9, non-physics 3, dead 5 / 6, suspicious 1. The 34 per region dropped as **weak** are the synergy candidates.

Variants on the table

1. **Quality-only pool.** Drop only HTmiss-linked, non-physics, dead and suspicious features; keep the weak ones; one feature per redundant block, the rest logged (1.23)
2. **Tokenize every new feature** through Gate A and the occupancy floor with shared edges; sentinels become missing tokens (F9). Gate A cannot score weak features, so the label-free tokenization cost decides
3. **Endcap first.** Wide-pool discovery with trees and transformers; count models only on small sets; the barrel checks (1.14–1.16)
4. **Gate each added feature or block** with class-conditional permutation: a drop is a detection; no drop means only "no effect above ε", with ε from a power study (1.20, F18)
5. **Risk checks before any claim:** data/MC audit, sample-ID test, data-vs-MC classifier, systematic variations, an η- and pileup-sliced repeat (F13–F17)
6. **Firewall at use, not at discovery.** Anything that reaches an ABCD axis or a network input uses one network's features only (D6, D7)

Open: which base the pool is rebuilt on (F10), and an operational "helps the grammar" metric before any test can say "helps".

Foundations: tokens, code and checks every goal relies on (1 of 3)

ID	Experiment	Source	Status
F1	**mk8 code of record** Tests for the six mk7 tools, closure verdicts with enrichment, a raw-variable axis, rebuild on fold seeds 1 and 2, one runner per registered prediction	mk8	Next
F2	**Review and test runs** Independent review of mk2 and mk7; run the 121-test mk2 v4 suite on your machine (110 passed on v3)	Vet V5, Prog §10	Open check
F3	**Endcap seed stability** Repeat the endcap token models on 3 training seeds (barrel done)	Vet V1, V16	Open check
F4	**Endcap cost per scheme** Tokenization cost at 8 and 16 bins and with Bayesian Blocks, including the transformer (32 bins is done)	Vet V2, V16	Open check
F5	**Bayesian Blocks as an axis** T2 edges through every model and the endcap axis comparison; needs its own plan after P1–P5	Vet A3, Prog §3.6, mk8 §1	Gated
F6	**Tail rule for quantile bins** Split the top and bottom bin at background 95% and 99%, screen as a new scheme (disclosed: proposed after seeing signal)	Prog D5	Needs decision
F7	**Model capacity** Endcap transformer without the epoch cap (300 epochs); a tuned barrel transformer	Prog D4, Vet V13, A7	Needs decision

Foundations: tokens, code and checks every goal relies on (2 of 3)

ID	Experiment	Source	Status
F8	**Unscreened tokenization paths** 15 of 19 viable paths never screened; the S, V and A axes never varied	Prog §2	Later
F9	**One shared tokenization** Edges fitted once per feature and region, so a feature has the same tokens in every set; the ~99000 sentinel becomes its own token	Add4 §2, §4	Proposed
F10	**Which base, which pool ranking** R0–R15 used base_mk1; Addendum 4 prescribes the new-base folds with the KS pool ranking recomputed on them	Prog header, Add4 §3	Needs decision
F11	**Refined-round comparators** Barrel comparison against the refined network rounds where they share LEXICON's folds	Vet V14, Page	Next
F12	**Closure bookkeeping** Check the nc definition in the closure code, then add the closure-as-systematic figure of merit next to the enrichment one; which base jobs exist at seeds 1 and 2	Vet V11, V12, §B.4; Add4 §10	Open check
F13	**Data/MC token audit** hits_ntot, nTracklet and every signal-only tail token: physics or bookkeeping? Required before any signal-side claim	Vet V4, Page	Next

Foundations: tokens, code and checks every goal relies on (3 of 3)

ID	Experiment	Source	Status
F14	**HTmiss leak gate** Flag any token with single-token AUC ≥ 0.995 or missing-pattern AUC ≥ 0.95; HTmiss-linked features excluded everywhere	mk2 §7 P1, Add4 §2	Proposed
F15	**Sample-ID and provenance tests** Within one class, can the features tell which MC sample or campaign an event came from? Audit for truth-derived inputs	FS §8	Proposed
F16	**Data-vs-MC classifier test** In a signal-depleted control region, can a classifier tell data from MC on the joint feature set?	FS §8	Proposed
F17	**Systematic variations** Rerun the evaluation on up/down systematic variations; a gain that disappears is fragile	FS §8	Proposed
F18	**Test power** An injection or bootstrap-in-N study that fixes ε before any “no effect” verdict	FS §7	Proposed
F19	**Barrel loose ends** Check the three-way MS-segment relation against the feature definitions; per-feature AUC on the barrel T2 run	Prog §3.5, §9	Open check

Goal 1 · Is there a grammar? (1 of 3)

ID	Experiment	Source	Status
1.1	**Exact vs pseudo-likelihood** MADE, causal and any-order transformer, GRU vs the current transformer, same folds, per network; scored on held-out fit, then dCor and closure; the exact model becomes the default if it closes better. Region order unsettled: page and Addendum 4 say barrel first, mk2 says endcap first	xNLL §3–5, mk2 3a-iii, Page, Vet A1	Next
1.2	**Order ensemble, per-order fits** Average several random orders (spread = per-event uncertainty) and keep each order's held-out likelihood: does any order beat seed noise?	xNLL §3.1, Page, mk2 1b	Next
1.3	**Order search** Greedy orders built from the any-order model's conditionals	Page	Gated
1.4	**Grammar budget** Total correlation per network from exact Chow–Liu tree and count models (no training seed); the share pairs, triples and a tree capture	Page, mk2 1a	Next
1.5	**Attention test** The same transformer with attention fixed to uniform: does attention, not size, carry the structure?	Page, Add4 §6	Next
1.6	**Gates, GLU and LSTM** Per-feature and missing-token sigmoid gates, a GLU feed-forward; an LSTM only if the GRU shows something	xNLL §3–4	Later
1.7	**Token classes** Group tokens into classes that keep most of the pairwise information	Page	Next
1.8	**Event types** A latent class model: a few production regimes, tokens independent within each	Page	Next

Goal 1 · Is there a grammar? (2 of 3)

ID	Experiment	Source	Status
1.9	**Correlation explanation** Latent factors ranked by how much total correlation each explains, stacked into a hierarchy	Page	Next
1.10	**Attention slots** A few learned queries pool the tokens; each slot is a candidate unit, matched across models by its masking footprint, read against a uniform-attention control	Add4 §5.2, §6	Proposed
1.11	**One scorecard and gates** Fit, share explained, stability and compression for every model (page); plus a shuffled-column null (Addendum 4)	Page, Add4 §6	Next
1.12	**Feature-set study, barrel** Dependency graphs on the NN lists, GAMBIT 16/10, the graded pool and random sets, each with a raw-feature control: which edges hold in every set, which features explain others away	Add4 §5.1, §6, §8	Proposed
1.13	**Label-free context set** Add pool features greedily by the background structure they add, no labels used	Add4 §3	Proposed
1.14	**Wide-pool latent search, endcap** Latent models trained on every graded feature together. Addendum 4's graded pool keeps only separating features, so it should run on the quality-only pool (1.23)	Add4 §5.2, §8, SBX	Proposed
1.15	**Recovery test and library** Units found on the wide pool must reappear in smaller sets, seeds and folds above the null; survivors form a library tagged NN1-only, NN2-only or cross-net	Add4 §5.2–5.3	Proposed
1.16	**Endcap GAMBIT feature set** Repeat the feature-set study on GAMBIT's endcap partition once it exists	Add4 §8	Gated

Goal 1 · Is there a grammar? (3 of 3)

ID	Experiment	Source	Status
1.17	**Endcap → barrel pretraining** Pretrain on endcap background, fine-tune on barrel, with a region token, a barrel-from-scratch control and the validity tests (zero-shot, domain classifier, sign check)	mk2 §11 P-b	Proposed
1.18	**Other pretraining options** Background-then-labels (regularization only, record it); a larger unlabeled stream (blocked until blinding rules exist)	mk2 §11 P-a, P-c	Later
1.19	**Masking design** Mask rate tuned per network; replacements drawn from background marginals; a mask token distinct from missing	mk2 §6, D8–D9	Proposed
1.20	**Low-separation feature tests** Class-conditional permutation, permutation vs removal, conditional mutual information, η- and pileup-sliced repeats; any structure found must survive nuisance shifts	FS §5–6	Proposed
1.21	**Estimator precision** Which total-correlation estimator is least biased at a few hundred events per fold?	IT §9	Proposed
1.23	**Quality-only token pool** Drop features only for quality (HTmiss-linked, non-physics, dead, suspicious), never for weak separation: about 69 endcap and 64 barrel features instead of 26 and 30; each tokenized and gated by 1.20	SBX, FS §4	Proposed
1.22	**Sequence among latent factors** Is there an order among the latent factors or slot co-activations? Named as a question, not yet planned	Page open, Add4 §9	Later

Goal 2 · Does signal speak differently? (1 of 3)

ID	Experiment	Source	Status
2.1	**Signal token models (A2)** The same model ladder on signal, read through background's edges; mass points pooled or fine-tuned	Page	Next
2.2	**Signal occupancy** Where signal lands in background's vocabulary; rare-in-background tokens become the audit list	Page	Next
2.3	**Signal transformer** Fine-tuned from the background transformer; must beat the tree and the uniform-attention control	Page	Next
2.4	**Separate tree structures** Signal and background trees compared edge by edge, with bootstrap stability	Page	Next
2.5	**Grammar contrast** Pair by pair, more or less co-occurrence in signal, with a false-discovery control and new-fold replication	Page	Next
2.6	**Likelihood ratio (A3)** Signal over background density, split exactly into what tokens carry alone and what their relations add	Page, mk2 2a	Next
2.7	**Structure contrast** The ratio with each class on its own tree: what a change of structure carries beyond a change of strength	Page	Next

Goal 2 runs after the data/MC token audit (F13): signal is MC, background is data.

Goal 2 · Does signal speak differently? (2 of 3)

ID	Experiment	Source	Status
2.8	**Order against physics** Expected orders from geometry and the decay chain written down before any order is learned	Page	Next
2.9	**Decay-chain test** Is information screened off along the pre-registered chain in signal but not in background?	Page, mk2 1c	Next
2.10	**Transformer ratios** Approximate and exact transformer ratios, read against the exact tree ratio	Page	Next
2.11	**Token probes (Route B, mk3)** One-hot tokens into logistic regression; a linear probe on frozen background-model embeddings	Prog §8.1 C, Routes, Page	Next
2.12	**Fine-tuned token classifiers** Transformer with a classification head, pretrained vs random init; event vectors into a small network	Routes	Gated
2.13	**Attention pooling and maps** A learned query builds one vector per event, tested against averaging; attention maps only as a cross-check on the tree structure	Page	Next

Goal 2 runs after the data/MC token audit (F13): signal is MC, background is data.

Goal 2 · Does signal speak differently? (3 of 3)

ID	Experiment	Source	Status
2.14	**Label as the next token** Append the class as the final token and predict it: the discriminative reading of your 4 Nov goal	mk2 2b	Proposed
2.15	**BERT vs autoregressive** Same features, folds, seeds and size; compared only on downstream results, both with a uniform-attention control	mk2 2d	Proposed
2.16	**Tokens into the networks (C)** Token or pooled event embeddings, or network-only library units, as extra NN1/NN2 inputs within each network's features	Page, mk2 2c, Add4 §5.4	Next
2.17	**Signal-side pretraining** Pretrain on all mass points, fine-tune per mass (carries the MC-vs-data caveat)	mk2 §11 P-d	Proposed
2.18	**Signed surprise** Per-feature surprises with signs fitted on labels: a diagnostic, not an axis	Prog §8.1 B	Later
2.19	**Which units signal breaks** Signal-vs-background shift for each library unit	Add4 §5.4	Gated

Goal 2 runs after the data/MC token audit (F13): signal is MC, background is data.

INVENTORY

Goal 3a · A token score as an ABCD axis (1 of 2)

ID	Experiment	Source	Status
3a.1	**P1** Endcap NN1 × S_2 (6 NN2 features, no timing, no hits_ntot), fold seeds 1 and 2	Vet §C	Registered
3a.2	**P2** Endcap NN1 × raw nTracklet: does tokenization add anything over the raw variable?	Vet §C	Registered
3a.3	**P3** Endcap NN2 × S_1 (NN1's 16 features); expectation stated: not viable	Vet §C	Registered
3a.4	**P4** Barrel NN1 × S_2 without timing, only after the barrel mass-point scan	Vet §C	Registered
3a.5	**P5** Label-free S_1 × S_2 in both regions: separation and closure	Vet §C	Registered
3a.6	**Barrel mass-point scan** Does the barrel have a shared bit like the endcap's? Gates P4; step 3 of the mk8 run order	Vet V8, mk8 §3	Next
3a.7	**Shared-bit checks** Confirm the bit on raw values (cluster energy 0, sentinel time); does timing carry coupling beyond it?	Vet V9, V10	Open check

INVENTORY

Goal 3a · A token score as an ABCD axis (2 of 2)

ID	Experiment	Source	Status
3a.8	**Attribution of survivors** Per-feature or per-block attribution and an audit for every variant that passes	Vet G11	Gated
3a.9	**Block surprise (A′)** Score whole blocks given the rest, blocks from background correlation, to remove the redundancy count	Prog D2–D3, Vet V3, A2	Needs decision
3a.10	**Direction-preserving combination** Two-sided per-feature p-values combined (Fisher, Stouffer, max) so directions stop cancelling	Vet A5	Proposed
3a.11	**Event-vector anomaly scores** Distance-based scores (nearest neighbours, Mahalanobis); needs its own plan after P1–P5	Vet A6, mk8 §1	Gated
3a.12	**Per-unit surprise** One surprise per library unit instead of one sum; label-aware weights count as Route B	Add4 §5.4	Gated
3a.13	**Ratio as an axis** The per-network likelihood ratio tested for closure like any network score (supervised), against a raw-feature control at equal AUC	Page, mk2 §8	Next
3a.14	**Token classifier as an axis** A supervised token classifier replaces NN1 or NN2 and goes through the same dCor and closure checks	Routes	Gated

INVENTORY

Goal 3b · Which features each network should get

ID	Experiment	Source	Status
3b.1	**Partition scoring** Sum the token dependence crossing the NN1/NN2 split: does it track closure across the shipped lists, GAMBIT's and random partitions? First check: GAMBIT barrel, where closure improved but dCor got worse. Does it beat output dCor as a predictor?	Add4 §7, IT §9	Proposed
3b.2	**Cross-net units** The same question with library units that span both networks	Add4 §5.4	Gated
3b.3	**Between-net dependence across jobs** Does input co-occurrence between the nets predict the NN pair's non-closure job by job? Needs its own plan after P1–P5	Vet A9, mk8 §1	Gated
3b.4	**Propose a split** Minimize shared information between the two input blocks while each keeps its separation; on the current sets, the top-40 pool, or both compared	mk2 3b-i, D6	Proposed
3b.5	**Check it against GAMBIT** Agreement scores per region; disagreements settled by closure under the BASELINE run spec, on folds disjoint from GAMBIT's seeds	mk2 3b-ii, GAM §2	Proposed
3b.6	**Splits in detector roles** Express both splits by detector role so barrel and endcap compare directly	GAM §3	Proposed
3b.7	**Inversion L1 → L2 → L3** From a healthy plane back to features: surrogate search, unit assignment, a factorized embedding, then plain NNs retrained. Waits on partition scoring being validated and the library	Add4 §10, mk2 3b-iii	Gated

BARREL + ENDCAP

An event as a set of tokens

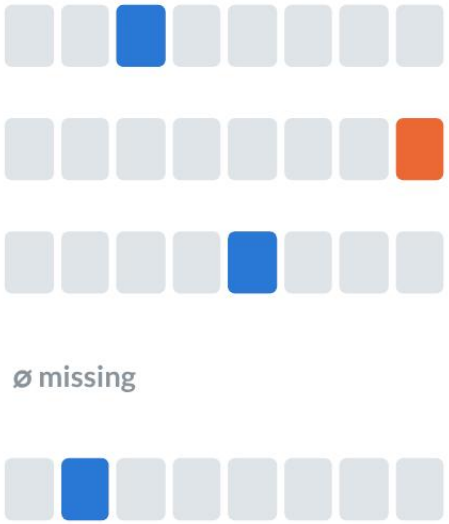
Continuous features

- nTracklet
- hits_ntot
- MSTracklet avg dR
- clusE
- MSeg dR

… 26 endcap, 30 barrel inputs

→ edges fitted on background only, per fold

Levels (schematic)



→

The event

{ nTracklet·3 | hits_ntot·top | MSTrk dR·5 | clusE·ø | MSeg dR·2 | … }

- A **set**, not a sentence: feature-ID embeddings, no positions. Any order is something to measure, not assume
- The missing / sentinel value is its own token, never a number
- Edges: equal-mass quantiles (T0_b*n*) or Bayesian Blocks (T2), screened by Gate A (information kept vs raw) and an occupancy floor
- Wave 1: **barrel T0_b8** (8 levels), **endcap T0_b32** (32 levels); bins under 5 background events merge into a neighbour

"You shall know a word by the company it keeps" (Firth): physics writes the grammar; the detector sets the alphabet.

What “grammar” means here: an amount and a form

1 Amount: total correlation

$$C(T) = \sum_i H(t_i) - H(t_1, \ldots, t_n) = D_{\mathrm{KL}}\left(p(t) \,\middle\|\, \prod_i p(t_i)\right)$$

Estimated on held-out background as the gap between an exact model and the independence model:

$$\widehat{C} = \frac{1}{N} \sum_{k=1}^{N} \left[\log p_{\mathrm{AR}}\left(t^{(k)}\right) - \sum_i \log p\left(t_i^{(k)}\right) \right]$$

$\widehat{C} > 0$ beyond seed spread is the minimum evidence of a grammar. It needs an **exact** likelihood: the chain rule holds for any feature order.

2 Form: hidden structure

Token classes, event types or factors that, once known, leave the tokens nearly independent:

$$p(t) = \sum_z p(z) \prod_i p(t_i \mid z)$$

- Scored on **one card**: held-out fit, share of C explained, stability across seeds, folds, tokenizations and feature sets
- Called a **grammar** only if the structure is hierarchical or follows the decay chain

A model that measures the amount without passing the latent gates is a density, not a grammar.

Pseudo-likelihood $\mathrm{PLL}(t) = \sum_i \log p(t_i \mid t_{\setminus i})$, the masked model's score so far, is **not** normalized: it cannot give H, I or C, and it undercounts features that predict each other.

Why a weakly separating feature can still matter

1 The chain rule

$$I(Y; X, Z) = I(Y; X) + I(Y; Z \mid X)$$

A sig/bg ranking (AUC, KS) measures $I(Y; Z)$. What Z adds is $I(Y; Z \mid X)$, which can be large when $I(Y; Z) = 0$.

XOR: X_1, X_2 symmetric about 0 and $Y = \text{sign}(X_1 X_2)$ give $I(Y; X_1) = I(Y; X_2) = 0$ and $I(Y; X_1, X_2) = 1$ bit.

2 Three ways it helps

- **Synergy**: information only in the joint distribution
- **Context**: barely separates, but changes what other features mean (η, pileup, barrel vs endcap)
- **Suppressor**: tracks the nuisance part of a good feature

Function words (*the*, *of*, *which*) cannot tell topics apart, and they **are** the syntax. Grammar lives in $p(x)$, not $p(y \mid x)$.

3 How it gets tested

- **Class-conditional permutation**: shuffle Z within each class; a drop is interaction, not marginal
- Held-out Δ log-loss of (X, Z) vs X, cross-fitted
- Repeat on η- or pileup-sliced samples

"Helps" can be shown. "Won't help" cannot, only "no effect above ε", with ε reported.

Consequence for LEXICON: selecting features by sig/bg separation removes context and synergy first, so the NN lists are biased against what a grammar search looks for. Discover on the **widest clean pool**; apply the per-net firewall only where tokens are **used**.

PROPOSED

The goals in detail: what each uses and delivers

G1 · no labels

Background grammar

Do background tokens have stable structure beyond independence: how much, in what form, in what order? Stable across seeds, folds, tokenizations and feature sets?

Tests: $\widehat{C}$ from exact models · latent units against a shuffled-column null · does any feature order beat seed noise

Delivers total-correlation budget · validated latent units · per-order likelihoods

AR amount, order OTHER form

G2 · uses labels

Signal against background

Does signal's structure differ from background's (edges, decay chain), and do tokens separate signal from background beyond the raw values?

Tests: tree edges compared class by class · decay-chain conditional MI · $\Lambda = \log p_S - \log p_B$ split into marginal + relational terms

Delivers likelihood ratio, marginal + relational · token classifiers · tokens as NN inputs · BERT vs AR

AR ratio MLM+AR classifiers MLM inputs

G3 · the ABCD plane

Axes and partitions

3a Does a background-only token score built from one net's features make a better axis than that net?
3b Does token dependence across the NN1/NN2 inputs predict non-closure, and which split does it propose against GAMBIT's?

Delivers per-net surprise axes · $E_\times$ per partition · a proposed split; later, inversion

MLM+AR 3a AR 3b

G1 is the precondition: if held-out $\widehat{C} \approx 0$, G2 and G3 reduce to per-feature effects and the language framing is empty. Every G2 item carries the MC-vs-data caveat; G1 and G3a token models never see a label (a G3a axis takes only a one-bit orientation sign).

PROPOSED

Crosswalk: where every earlier framing lands

Proposed goal	Public page (30 Oct)	Addendum 4 (2 Nov)	Goal set mk2 (4 Nov)	Steve's goals (4 Nov)	Routes
G1 Background grammar	Verdict 1 · Q2	V1	1a total correlation · 1b order · 1d latent	–	A1 models, latent grammar, A1 variant
G2 Signal vs background	Verdict 2 · Q3 · Q1 *(no verdict)*	V2	1c decay chain · 1e A2 · 2a–2d	Goal 2: next token → sig / bkg	A2, A3, B, C
G3a Token axis	Verdict 3 *(no question)*	V3	3a-i to 3a-iii	–	A1 per net
G3b Partitions	–	V4 · inversion §10	3b-i to 3b-iii	Goal 1: what GAMBIT does	Partition scoring; GAMBIT cross-check

What changes on the public page: one question per goal (the page's Q1 has no verdict and verdict 3 has no question); Q1 survives as the raw-feature control every goal already needs.

What changes from mk2: only 1c and 1e move to G2. **From Addendum 4**: nothing dropped; V4 plus inversion is G3b; Routes B and C, listed there as supervised variants of V3, move to G2 as in mk2.

Two model families, and they measure different things

	Objective	Gives	Cannot give	In LEXICON
MLM BERT-like	$\sum_i \log p_\theta(t_i \mid t_{\setminus i})$	Context-rich embeddings; cheap surprise	A normalized density; exact H, I, C, likelihood ratio	CutFormer; pair / triple count models (pseudo)
AR chain rule	$\sum_i \log p_\theta\left(t_{\pi(i)} \mid t_{\pi(<i)}\right)$	Exact likelihood for any order π; entropy, MI, Neyman–Pearson ratio	An order-free representation	MADE, GRU density, any-order CutFormer, Chow–Liu tree
OTHER	latent variable · skip-gram	Own likelihoods / embeddings	–	Event types, CorEx; Evt2Vec

1 **AR carries the measurements.** Size of the grammar, order, decay chain, likelihood ratio, partitions all need exact likelihoods.

2 **The run order is inverted.** The MLM has run; no AR model has. AR vs PLL on the same folds unlocks G1, G2's ratio and G3b.

3 **No shared intrinsic metric.** BERT vs AR is compared downstream only (closure, fine-tuned AUC, gain as NN inputs), always with a uniform-attention control.

Rules that hold for every goal

Stage	Vocabulary	Per-net firewall (D6)
Discovery: G1, G2 structure, G3b graphs	Widest clean pool, both nets' features	not applied
Use as an ABCD axis (G3a)	One net's features only	**applied**
Use as NN inputs or token classifiers	One net's features only	**applied**

Structure found on the wide pool enters an axis only through units whose features all belong to that net. Disjoint lists are not enough: separation is **measured** ($dCor^2 < 0.01$, mass-point scan).

- **HTmiss split by construction.** Background < 40 GeV, signal ≥ 40 GeV: HTmiss-linked features are excluded everywhere
- **Signal is MC, background is data.** Signal-only tokens are audited before any G2 claim
- **Discovery trains on background only.** Labels enter only through the choice of vocabulary, which is recorded
- **Barrel is small.** The wide-pool latent search runs in the endcap and the barrel checks it; the exact-likelihood variant keeps its barrel-first order
- **Thresholds before runs**, comparisons paired by seed, results never deleted, only superseded

Masking is not augmentation. Every masked input is a function of the same N events, so N_{eff} is unchanged; masking only raises targets per event. New information comes only from new events (endcap → barrel pretraining) or a real symmetry.

PART II

What has run

R0–R15, 12–19 Oct 2026, on base_mk1 · every result here is exploratory

The pipeline, steps 0 to 5

0 — Freeze edges
per fold, background only

1 — Gate A
tokens vs raw, cross-fitted

2 — Occupancy
merge-to-floor

3 — Token models
independence, counts, CutFormer, raw control, Evt2Vec

4 — dCor(surprise, NN)
per-net, independence + counts

5 — Closure
surprise as one ABCD axis

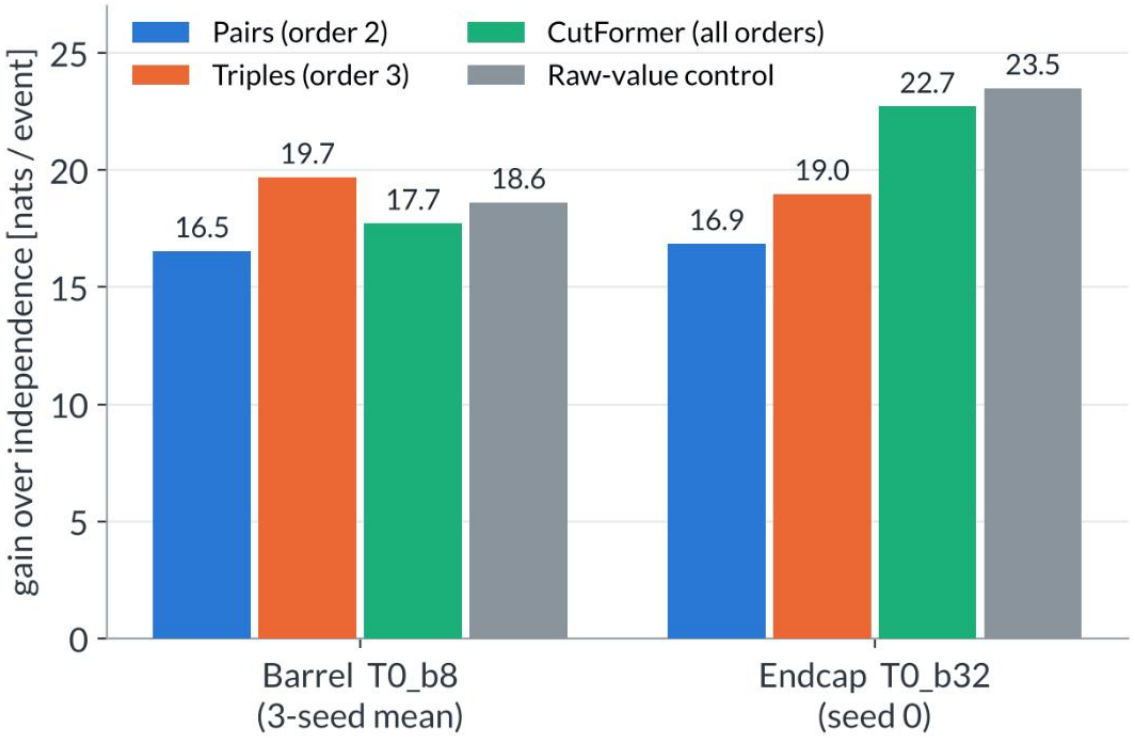
Screen v5, wave 1	Gate A	Verdict
Endcap · T0_b32	25 GOOD, 1 flagged, 0 BAD	**PASS**
Barrel · T0_b8	13 GOOD, 16 INDETERMINATE, 0 BAD	**PASS**

Coverage: 120 path options → 25 → 19 viable. 8 screened (4 schemes × 2 regions); step 3 trained on the endcap wave-1 path and all 4 barrel schemes. 15 viable paths never screened; the S, V and A axes never varied. Bayesian Blocks (T2) was dominated by T0 in both regions.

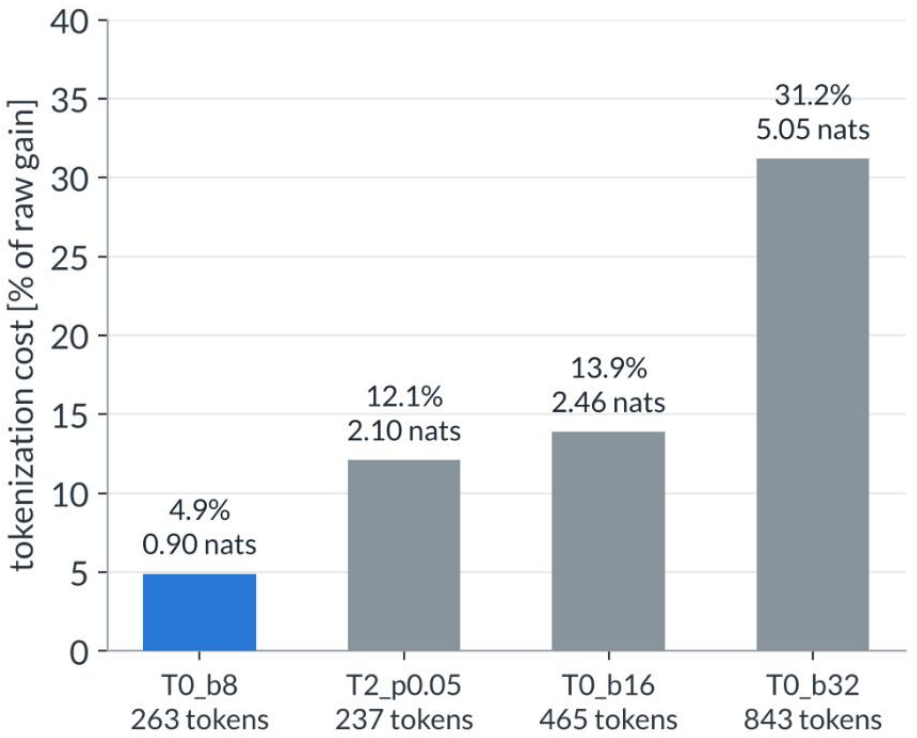
Data: base_mk1 Round_1, base seed 0, 5 folds, scored out of fold · barrel 1,089 bkg (≈609 training per fold) + 4,987 sig · endcap 34,970 bkg (≈23,780 per fold) + 9,996 sig

Tokens predict each other, and tokenizing is cheap

A. Gain over independence, held-out background



B. Barrel: what tokenizing loses, by scheme



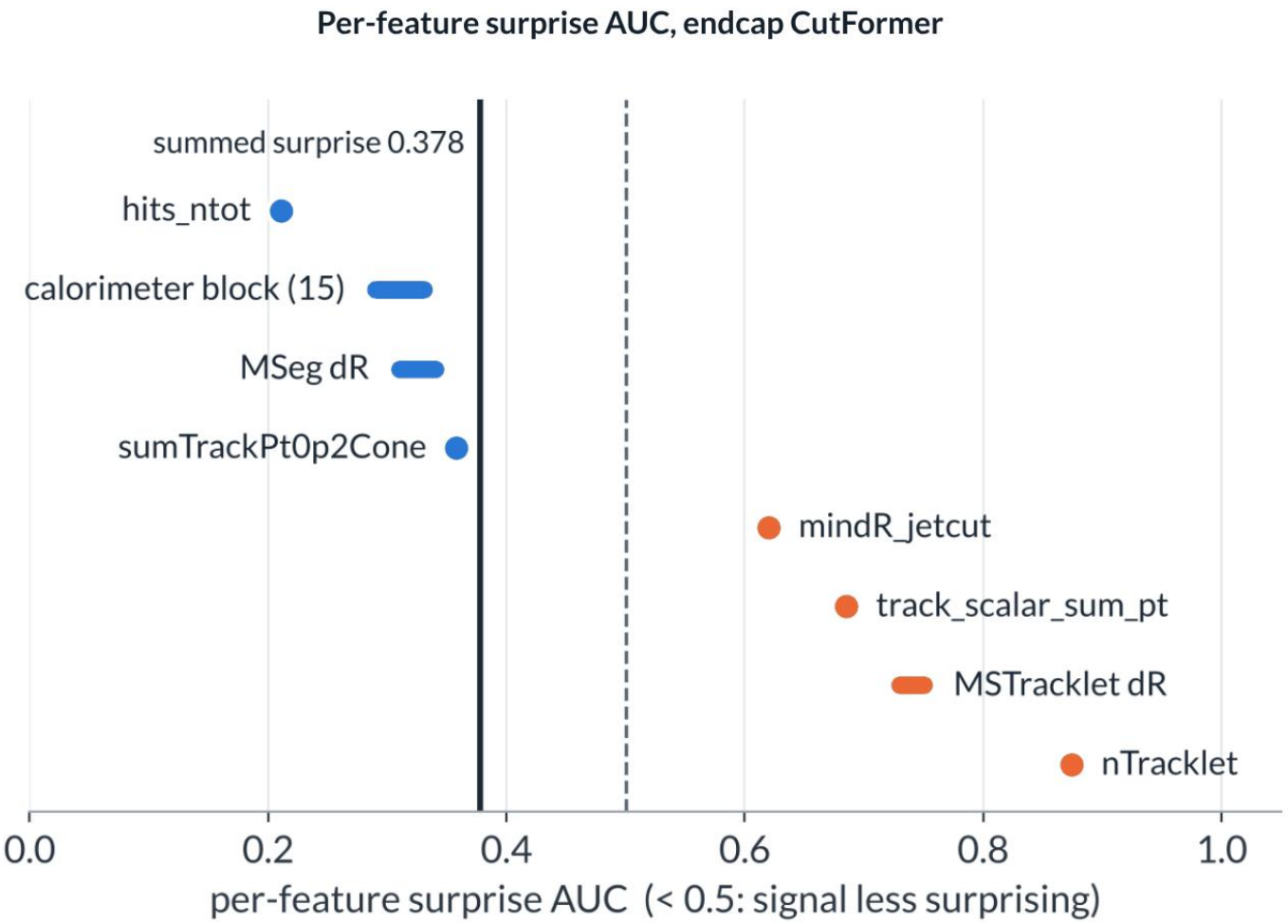
1. Co-occurrence is large: at least **+19.7** (barrel, triples) and **+22.7** (endcap, CutFormer) nats/event; mostly pairwise (order 2 holds 84% / 89% of order 3)

2. Barrel: triples beat the transformer on every seed (609 events per fold underfit it). Endcap: CutFormer goes 3.7 nats past triples

3. At 8 bins tokenizing costs 4.9% of the raw gain; at 32 bins 31%: 843 tokens, more than the 740 training events they are fitted on

Data: Held-out background pseudo-log-likelihood, nats/event, out of fold · barrel T0_b8: training seeds 0, 1, 2 (mean shown); endcap T0_b32: seed 0 only · panel B: seed 0 except T0_b8 (3-seed mean) · base_mk1 Round_1, base seed 0, 5 folds, scored out of fold · barrel 1,089 bkg (≈609 training per fold) + 4,987 sig · endcap 34,970 bkg (≈23,780 per fold) + 9,996 sig

Summed surprise points the wrong way



1 Directions cancel

Single features reach 0.21 and 0.87, pointing opposite ways. The sum lands at **0.378**, below a coin flip.

2 Redundancy is counted 15 times

The calorimeter features predict each other. Background pays at each of the 15; signal sits on shared zero atoms. Without the block, surprise reaches **0.685**.

3 Open top bins hide the tail

Barrel T0_b8: **85%** of signal `hits_ntot` lands in the top token, which holds 17% of background and is its cheapest. Bayesian Blocks split it and flip the direction. No background-only check can see this.

Data: Per-feature AUC of CutFormer surprise, all signal mass points pooled, out of fold, seed 0 · open-top-bin numbers: barrel fold 0, 999 signal / 218 background events · base_mk1 Round_1, base seed 0, 5 folds, scored out of fold · barrel 1,089 bkg (≈609 training per fold) + 4,987 sig · endcap 34,970 bkg (≈23,780 per fold) + 9,996 sig

The first axes leaked: per-net surprise (D6)

Before (R8–R10): one surprise from every feature

NN1 inputs
16 endcap · 20 barrel

NN2 inputs
10 endcap · 10 barrel

↓ both feed one score ↓

Surprise S shares **62–67%** of its inputs with NN1 and 33–38% with NN2

Paired with either net, the plane is not two independent views. Endcap NN1 × S: dCor² 0.21–0.23, ten times the vanilla NN pair.

> What it looked like: barrel NN2 × T2 surprise closed in 25–33 of 49 jobs, but kept only about a third of the signal in A; the closing NN pairs kept 83–88%.

After (R11 on): each axis from the other net's features

NN1 inputs → S_1

NN2 inputs → S_2

NN2 × S_1

NN1 × S_2

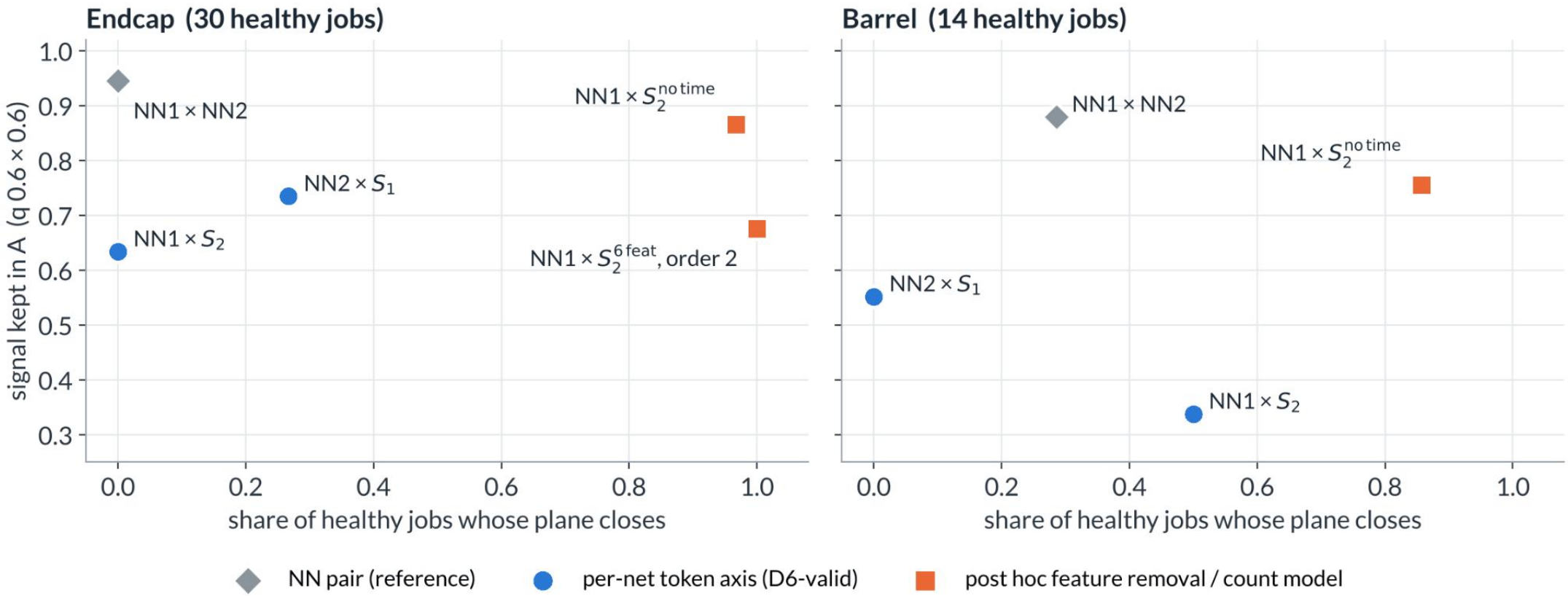
$S_1 \times S_2$ label-free

Every learned part is fitted on that subset only. R8–R10 stay in the record, relabelled as the leaky baseline; the barrel verdict "Route A is dominated" was narrowed to summed, all-feature surprise.

> Separation is also **measured**, not assumed from the lists: a cross-side mass-point scan, and dCor² < 0.01 for the label-free plane (G6).

Data: 49 base-seed-0 jobs (vanilla, DisCo, closure-regularized); healthy = NN1 AUC > 0.8 and NN2 AUC > 0.9: 30 endcap, 14 barrel · closes = |nc|med ≤ shuffle-null p95 · signal in A at bkg quantile 0.6 × 0.6

Separated axes close where the NN pair does not



1 Endcap NN2 × S_1 closes in **8 of 30** healthy jobs; the NN pair in 0 of 30. It keeps 0.735 of signal in A against 0.945

2 Endcap NN1 × S_2 without the 3 timing features: **29 of 30**, signal in A 0.866. Found after seeing results

3 The upper right corner is where an axis would beat the NN pair outright; nothing confirmed sits there yet

Data: 49 base-seed-0 jobs (vanilla, DisCo, closure-regularized); healthy = NN1 AUC > 0.8 and NN2 AUC > 0.9: 30 endcap, 14 barrel · closes = |nc|med ≤ shuffle-null p95 · signal in A at bkg quantile 0.6 × 0.6

One shared bit, and density is not axis quality

1 One shared bit: "the vertex has no calorimeter cluster" (D7)

34.7%
of background on the shared default token

68.2%
of signal on the same token

It is carried by `clusE`-type features (NN1) **and** `clustime` features (NN2). S_1 vs timing surprise: Spearman **0.70**, falling to **0.035** once that token is excluded.

Label-free $S_1 \times S_2$: dCor² 0.67 → 0.0005 with the 3 timing features removed. The NN1/NN2 split itself shares this bit at the input level.

2 Count models improve the density, not the axis

Endcap, per-net	Order 1 → 2 → 3
S_1 PLL (nats)	39.73 → 29.27 → 27.89
NN2 × S_1 closes	8 → 9 → 8 of 30
S_2 PLL (nats)	13.01 → 9.35 → 9.35
NN1 × S_2 closes (order 1 → 2)	29 → 30 of 30
… signal in A (order 1 → 2)	0.876 → 0.676

Conditioning re-weights what the summed surprise responds to. Background likelihood and axis quality are different objectives.

Data: 49 base-seed-0 jobs (vanilla, DisCo, closure-regularized); healthy = NN1 AUC > 0.8 and NN2 AUC > 0.9: 30 endcap, 14 barrel · closes = |nc|med ≤ shuffle-null p95 · signal in A at bkg quantile 0.6 × 0.6 · count models: Dirichlet backoff, 3-SE validation guard

Where the evidence stands

Holds so far

- Co-occurrence among tokens is large and mostly pairwise, in both regions
- Tokenizing keeps 94–97% of the context information at the chosen bin counts
- Summed surprise cancels signal information; the cause is the scoring
- Bin edges in the tail decide where signal lands
- The leak ran both ways (D6); disjoint inputs can still share a bit (D7)
- A better background density did not give a better axis

Not yet a measurement

- One fold seed (base seed 0); first-round sweep, not the refined models
- The best plane came from feature removals chosen after seeing results
- `nTracklet` carries most of S_2's signal: the raw-variable control is not run
- Data/MC audit of `hits_ntot`, `nTracklet` and signal-only tokens is open
- No exact-likelihood (AR) model has run, so G1's amount is not yet measured
- mk7 tools were tested on placeholder arrays only, no independent review

PART III

Plan

Confirm what is registered, then run what each goal needs

BARREL + ENDCAP

Registered predictions (vetting v1, 19 Oct)

[redacted]	NN2 × S_1	Endcap	F1 and F4. Stated expectation: not viable
	NN1 × S_2	Barrel	Only after the barrel mass-point scan passes; F1 and F4 against the same job's NN pair
	S_1 × S_2	Both	dCor² < 0.01 and $r \leq 1$; signal in A reported descriptively

F1 gate: the plane closes in ≥ $\rho = 0.8$ of healthy jobs on each fold seed. **F4:** median enrichment ratio $e_L / e_{\text{ref}} \geq \eta = 0.8$ over closing jobs, against the same job's NN pair.

Closure distance $r = |nc|_{\text{med}} / p_{95}^{\text{null}}$, $r \leq 1$ closes. These test **G3a** only; G1, G2 and G3b have no registered run yet.

PROPOSED

What each goal needs, in order

1. **mk8, code of record (F1).** Tests for the six mk7 tools, enrichment and F1/F4 verdicts, raw-variable axis, fold-seed rebuild, one runner per prediction — G3a
2. **P1–P5 (3a.1–3a.5)** on fold seeds 1 and 2, both regions; barrel mass-point scan (3a.6) first — G3a
3. **Exact likelihood vs PLL on the same folds (1.1, 1.2, 1.4).** First AR model; gives $\widehat{C}$, per-order likelihoods, block MI — G1 G2 G3b
4. **Feature-set study, barrel (1.12, 3b.1).** One shared tokenization; NN lists, GAMBIT 16/10, graded pool, random sets; $E_\times$ against GAMBIT's barrel result — G1 G3b
5. **Wide-pool latent search, endcap (1.14, 1.15),** with a shuffled-column null; recovery of units across feature sets, seeds and folds — G1
6. **Data/MC token audit (F13),** then signal models, likelihood ratio, token classifiers, BERT vs AR (2.1–2.16) — G2
7. **Later:** endcap → barrel token pretraining (1.17); inversion (3b.7) from a healthy plane back to features, gated on $E_\times$ tracking closure — G1 G3b

Open decisions

1. **Goal spine.** Adopt G1 / G2 / G3a / G3b for the public page and the progress document, or keep mk2 as written?

2. **Run order.** Promote exact likelihood vs PLL ahead of the feature-set study?

3. **AR order policy.** Greedy by information, random-order ensemble, or learned?

4. **Data base.** R0–R15 ran on base_mk1; Addendum 4 assumes the new-base files. Which base do the next rounds use, and is the KS pool ranking recomputed on its folds?

5. **Wide pool.** The SIG/BG graded pass set, or a fixed top-N?

6. **Still awaiting an answer:** D1 (rule for INDETERMINATE features), D2 (next build) and D5 (tail rule for T0)

Decisions that set how the experiments run

Exact-likelihood setup (1.1–1.2): ensemble size K; average p or $\log p$ over orders; GRU in the barrel or not; which model is the primary comparison; barrel or endcap first

Order policy (1.2–1.3): greedy by information, a random-order ensemble, or learned

Wide pool and bins (1.12–1.15): graded pass set or fixed top-N; one bin level per region, or the endcap at 8 bins for parity; how the label-free set is selected

Fixed in advance (1.10, 1.15): slot count K, number of random-set draws, recovery threshold, the cross-net tag threshold

Axis combination (3a.12): label-blind per-unit surprises only, or also a label-aware variant reported as Route B

Inversion (3b.7): weights α, λ, μ fixed or swept; what separation cost buys a given closure gain

Masking and pretraining (1.17–1.19): mask rate per network or the BERT default; special tokens; adopt endcap → barrel pretraining now

"Helps the grammar" (1.20): an operational metric is needed before any feature test can say "helps"